EASY · ADVANCED MATH · ANSWER KEY

Advanced Math

14

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1

A

A. The estimated value of the tablet was $\$ 225$ when it was purchased.

Choice A is correct. It's given that the graph shown gives the estimated value y, in dollars, of a tablet as a function of the number of months since it was purchased, x. The y-intercept of a graph is the point at which the graph intersects the y-axis, or when x is 0. The graph shown intersects the y-axis at the point $0, 225$. It follows that 0 months after the tablet was purchased, or when the tablet was purchased, the estimated value of the tablet was 225 dollars. Therefore, the best interpretation of the y-intercept is that the estimated value of the tablet was $\$ 225$ when it was purchased.

Choice B is incorrect. The estimated value of the tablet 24 months after it was purchased was $\$ 50$, not $\$ 225$.

Choice C is incorrect. The estimated value of the tablet had decreased by $\$ 225 - \$ 50$, or $\$ 175$, not $\$ 225$, in the 24 months after it was purchased.

Choice D is incorrect and may result from conceptual errors.

Q2

7

The correct answer is 7. It's given that the x-intercept of the graph shown is $x, 0$. The graph passes through the point $7, 0$. Therefore, the value of x is 7.

Q3

C

C. 36

Choice C is correct. It's given that $f x = 8\sqrt{x}$. Substituting 48 for $f x$ in this equation yields $48 = 8\sqrt{x}$. Dividing both sides of this equation by 8 yields $6 = \sqrt{x}$. This can be rewritten as $\sqrt{x} = 6$. Squaring both sides of this equation yields $x = 36$. Therefore, the value of x for which $f x = 48$ is 36.

Choice A is incorrect. If $x = 6$, $f x = 8\sqrt{6}$, not 48.

Choice B is incorrect. If $x = 8$, $f x = 8\sqrt{8}$, not 48.

Choice D is incorrect. If $x = 64$, $f x = 8\sqrt{64}$, which is equivalent to 64, not 48.

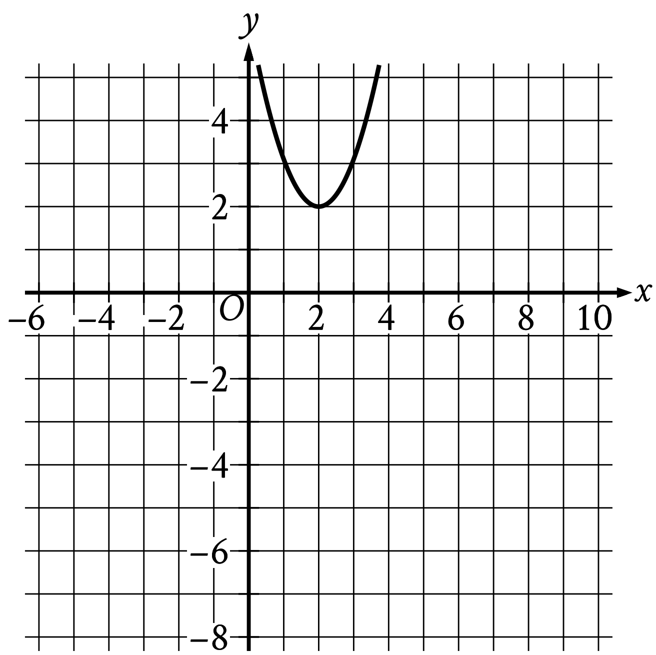
Q4

77

The correct answer is 77. It’s given that the function f is defined by $f x = x^2 + x + 71$. Substituting 2 for x in function f yields $f 2 = 2^2 + 2 + 71$, which is equivalent to $f 2 = 4 + 2 + 71$, or $f 2 = 77$. Therefore, the value of $f 2$ is 77.

Q5

A



A.

- The parabola opens upward.
- The vertex is at the point (2 comma 2).
- The parabola passes through the following points:
  - (1 comma StartFraction 31 Over 10 EndFraction)
  - (2 comma 2)
  - (3 comma StartFraction 31 Over 10 EndFraction)

Choice A is correct. When a graph is translated up 4 units, each point on the resulting graph is 4 units above the point on the original graph. In other words, the y-value of each point on the graph increases by 4. The graph shown passes through the points $1, -1$, $2, -2$, and $3, -1$. It follows that when the graph shown is translated up 4 units, the resulting graph will pass through the points $1, -1+4$, $2, -2+4$, and $3, -1+4$. These points are $1, 3$, $2, 2$, and $3, 3$, respectively. Of the given choices, only the graph in choice A passes through the points $1, 3$, $2, 2$, and $3, 3$.

Choice B is incorrect. This is the result of translating the graph down, rather than up, 4 units.

Choice C is incorrect. This is the result of translating the graph left, rather than up, 4 units.

Choice D is incorrect. This is the result of translating the graph right, rather than up, 4 units.

Q6

A

A. The object traveling at 34 meters per second has a kinetic energy of $5,202$ joules.

Choice A is correct. It's given that the kinetic energy, in joules, of an object with a mass of 9 kilograms traveling at a speed of v meters per second is given by the function K, where $K v = \frac{9}{2} v^2$. It follows that in the equation $K 34 = 5,202$, 34 is the value of v, or the speed of the object, in meters per second, and $5,202$ is the kinetic energy, in joules, of the object at that speed. Therefore, the best interpretation of $K 34 = 5,202$ in this context is the object traveling at 34 meters per second has a kinetic energy of $5,202$ joules.

Choice B is incorrect. The object traveling at 340 meters per second has a kinetic energy of $520,200$ joules.

Choice C is incorrect. The object traveling at $5,202$ meters per second has a kinetic energy of $121,773,618$ joules.

Choice D is incorrect. The object traveling at $23,409$ meters per second has a kinetic energy of $2,465,915,764.5$ joules.

Q7

D

D. $0, 8$

Choice D is correct. The y-intercept of a graph in the xy-plane is the point at which the graph crosses the y-axis. The graph shown crosses the y-axis at the point $0, 8$. Therefore, the y-intercept of the graph shown is $0, 8$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q8

31

The correct answer is 31. It's given that the y-intercept of the graph of $y = x^2 + 31$ in the xy-plane is $0, y$. Substituting 0 for x in the given equation yields $y = 0^2 + 31$, or $y = 31$. Thus, the value of y is 31.

Q9

C

C. 23

Choice C is correct. The value of $f2$ is the value of fx when $x = 2$. Substituting 2 for x in the given function yields $f2 = 2^3 + 15$, or $f2 = 8 + 15$, which is equivalent to $f2 = 23$. Therefore, the value of $f2$ is 23.

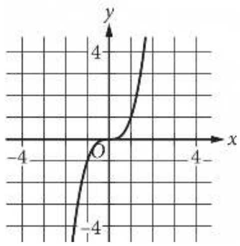
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $f2$ when $fx = x3 + 15$, rather than $fx = x^3 + 15$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q10 B

B.



Choice B is correct. Each pair of values shown in the table gives the ordered pair of coordinates for a point that lies on the graph that represents the relationship between x and y in the xy-plane: $(0,0)$, $(1,1)$, $(2,8)$, and $(3,27)$. Only the graph in choice B passes through the points listed in the table that are visible in the given choices.

Choices A, C, and D are incorrect. None of these graphs passes through the point $(1,1)$.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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